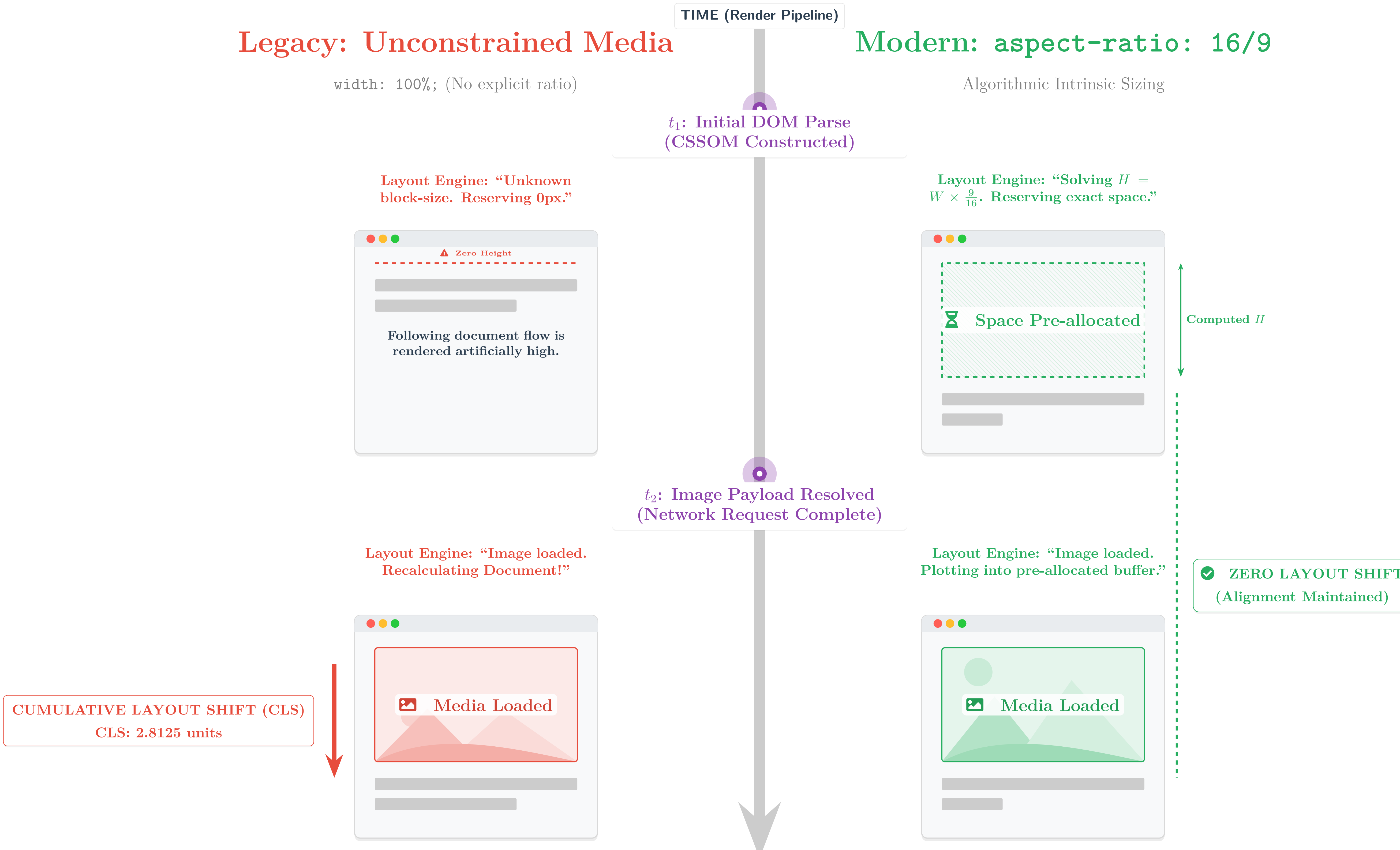
Temporal Layout Mechanics & CLS Mitigation

How `aspect-ratio` preempts layout thrashing chronologically during the render pipeline



Layout Thrashing

Without `aspect-ratio`, the browser cannot mathematically derive the height until the physical file's headers arrive over the network. It defers layout space (0px). When the payload finally arrives, it forces a computationally expensive geometry recalculation of the entire document tree, shifting all subsequent nodes and jarring the user.

Preemptive Geometric Resolution

The `aspect-ratio` specification acts as a declarative mathematical constraint. The layout engine computes the target bounding box instantly during the initial CSSOM render tree construction. The physical bytes of the image are strictly decoupled from the DOM geometry.